An Exact Conditional Null for Correlated Failure in Open-Model Ecosystems, and a Route to Anytime-Valid Monitoring

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Abstract

Measuring correlated failure between language models requires a null hypothesis for "independent enough," and prior work shows the conclusion is highly sensitive to which null is chosen, leaving the choice apparently subjective. We show one null is not a choice: the row and column margins of a model-by-item outcome matrix are the jointly sufficient statistics of the Rasch family, so the uniform distribution over margin-matched matrices is the unique fit-free conditional null for that family. At this null, on 1,228–1,362 open models across five benchmarks, the commonly reported mean-level "excess" co-failure is a deterministic function of item difficulty and vanishes identically, while a second-order residual correlation of 2.9–11.9$\times$ its null survives, is robust to removing 21–64% of models as near-duplicates, and concentrates in 4–12 latent dimensions rather than one shared factor. Our test is a fixed-sample exact randomization test, and because open leaderboards grow continuously the natural downstream use is a standing monitor. We build the fixed-set-of-looks version on the real arrival stream: twelve monthly looks over a population growing from 241 to 3,762 models, a Monte Carlo p-value against each look's own exact null, the standard calibrator, and the arithmetic mean of the calibrated e-values, which is valid under the severe dependence that nested populations induce. Against a control stream in which the null holds by construction the merged e-value stays at 0.90 (maximum 1.17 over all looks), as a valid e-value must; on real data every look saturates the Monte Carlo resolution ceiling, which we report as a ceiling rather than as unbounded evidence. What the exact null does not yet supply is an e-process permitting optional stopping, and we state precisely why: conditioning on the accumulated margins does not deliver a null for the increment when a row is added.

1 Motivation

If many deployed models share blind spots, an ensemble, a panel of model judges, or a market of competing screening vendors supplies far less independent evidence than its headcount suggests. A growing literature measures how often models err together (Kim et al., 2025), almost all of it by comparing the rate at which models fail the same items against a baseline in which failures are independent. Jo et al. (2026) show this comparison is highly sensitive to the assumed baseline: conditioning on item difficulty attenuates residual correlations, "some even flipping from strongly positive to slightly negative," and they prove that a sufficiently expressive null can explain away any observed dependence. Their stated conclusion is that the choice of null is subjective. This paper argues that the ladder of nulls has a canonical rung, reports the exact answer there, and builds the sequential monitor that answer supports.

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		rms $\lvert R_{ij}\rvert$			participation ratio			
Benchmark	N	obs.	null	ratio	obs.	null	ratio	dims > edge
ARC-Challenge	1362	0.2483	0.0449	5.5×	23.6	449.3 ± 3.4	0.052	6
Winogrande	1361	0.2691	0.0409	6.6×	20.1	493.8 ± 3.0	0.041	5
TruthfulQA	1334	0.2942	0.0583	5.1×	17.4	309.7 ± 2.4	0.056	5
GSM8K	1228	0.0987	0.0346	2.9×	118.6	553.1 ± 2.2	0.214	4
HellaSwag	1362	0.2457	0.0206	11.9×	27.3	958.0 ± 3.2	0.028	12

Table 1: Residual structure after exact-margin conditioning. Null rms is the mean over 40 curveball replicates; null PR is mean $\pm$ SD over 60 independent-chain replicates, each burned in for 50 trades per model with both margins exactly preserved. Uncalibrated, PR reads 1.5–3.6 on every benchmark, a number driven almost entirely by the shared item-difficulty eigenvalue that conditioning removes.

2 The canonical null

Let $F \in \{0,1\}^{N\times M}$ record $F_{im} = 1$ iff model i fails item m, with row sums r_i and column sums c_m. The Rasch model $\Pr(F_{im} = 1) = \sigma(\alpha_i + \beta_m)$ has $\{r_i\}$ and $\{c_m\}$ as jointly sufficient statistics, so the conditional law of F given its margins is uniform on the set of margin-matched binary matrices for every parameter value (Rasch, 1960; Andersen, 1973). That set is the ecologists' fixed-fixed null. We sample it with curveball (Strona et al., 2014), whose convergence to the uniform distribution on the fibre is proved by Carstens (2015) provided failed trades are counted as steps (ours does), and which we verify directly: enumerating every matrix on four small fibres (sizes 48, 90, 204, 1,170) and drawing 60,000 thinned samples, a χ^2 test against uniformity is not rejected on any ($p = 0.23, 0.05, 0.73, 0.15$); on the real matrices, four dispersed chains give Gelman–Rubin $\hat{R} \in [0.98, 1.03]$ at the burn-in used. Unlike a fitted item-response null, this null requires no optimizer, no penalty, and no identification convention: within the Rasch family it is the strongest ability-and-difficulty control obtainable without estimating anything.

Lemma 1 (Mean co-failure depends only on item margins)**.** *The mean pairwise co-failure rate satisfies* $O = \frac{1}{N(N-1)} \sum_{i\neq j} \frac{1}{M} \sum_m F_{im}F_{jm} = \frac{1}{MN(N-1)} \sum_m (c_m^2 - c_m)$.

Proof. $\sum_{i\neq j} F_{im}F_{jm} = (\sum_i F_{im})^2 - \sum_i F_{im}^2 = c_m^2 - c_m$ since $F_{im} \in \{0,1\}$. □

Any randomization preserving the item margins therefore leaves O invariant with exactly zero variance, so the "excess" of O over this null is identically zero by construction rather than empirically small. The same degeneracy is known in ecology as that of Schluter's V-ratio (Schluter, 1984; Gotelli, 2000), in psychometrics as Cronbach's α (Cronbach, 1951), and in inter-rater agreement as Fleiss's observed agreement (Fleiss, 1971); to our knowledge none of these is cited in the model-evaluation literature that reports the statistic, and the full paper states and attributes the identities.

3 What survives at the exact null

Statistics. Let P be the Rasch mean field matching the observed margins (margin error below 10^{-11}), a point estimate of the null mean of $FF^\top/M$ that we validate against the randomization itself: over 319,600 ARC model pairs the curveball expectation of $C_{ij} = \frac{1}{M}\sum_m F_{im}F_{jm}$ and the Rasch prediction correlate at 0.99999. With $D = (FF^\top - PP^\top)/M$ and R its unit-diagonal scaling, we report the rms off-diagonal $\lvert R_{ij}\rvert$, the participation ratio $\mathrm{PR} = (\sum_k \lambda_k^+)^2 / \sum_k (\lambda_k^+)^2$ of R's positive spectrum, and the number of eigenvalues of R above the null's largest (parallel analysis against the exact null rather than iid noise), each against exact-margin replicates.

Data. Per-item outcomes are harvested from the archived Open LLM Leaderboard (1,228–1,362 models per benchmark, five benchmarks) at zero inference cost by reading only the metric column of each result file. Item identity is resolved by row index, licensed by a zero-mismatch row-order audit over 334 models spanning the archive's two schema generations.

Results. Table 1 shows residual correlation at 2.9–$11.9\times$ its null and a participation ratio at 2.8–21.4% of its null on every benchmark. Neither number licenses the reading "1,362 models behave like 24": for the unclipped spectrum, $\mathrm{PR} = N/(1 + (N-1)\,\overline{R_{ij}^2})$ exactly (16.04 on ARC computed either way), so PR is a monotone transform of mean squared residual correlation and cannot separate one weak global factor from a partition into tight clusters. The diagnostics that can separate them say the structure is neither: on ARC the leading eigenvalue carries 54% of squared residual mass (a single factor would carry $\approx 99\%$, twenty exact clone blocks $\approx 10\%$), deflating it raises PR only to 46.8 rather than toward N, and six eigenvalues exceed the null's spectral edge of 27.9. A two-parameter item-response simulation with discrimination heterogeneity of $\mathrm{sd}(\log a) = 0.60$ produces rms residual correlation of 0.076 against the observed 0.248, so discrimination heterogeneity alone is not a sufficient explanation, while a simulated population with two ability dimensions and no shared lineage reproduces much of the effect ($\lambda_1^2/\sum \lambda_k^2 = 0.457$ against the observed 0.540).

Robustness. Deduplicating at agreement 0.95, which discards 21–64% of models, with the Rasch fit and null redrawn, moves the rms ratio by at most $1.14\times$ (ARC $5.58 \to 5.44$); the PR ratio moves more, by $2.44\times$ on HellaSwag and at most $1.46\times$ elsewhere. A later pass over the growing archive (2.2–$2.8\times$ this population) moved every PR ratio down, by 0.005–0.010 on four benchmarks and 0.023 on GSM8K. On the independently implemented successor harness (ARC, 212 models) the rms excess is $4.1\times$, agreeing in direction but not magnitude. On planted data the test reaches power 1.00 against shared failure modes at a planted rms of 0.048 and rejects at 0.017 under the null; against the alternative in which every model shares one item-difficulty profile, power is exactly α, because that alternative is the null by construction.

A refuted pre-registration. The pre-registered primary statistic was the dispersion $\mathrm{Var}_{i<j}(C_{ij})$. It came out below its null on three benchmarks and above on two, its kill condition fired, and we report it as refuted rather than reinterpreted. The second-order structure above is what survives, not the statistic we planned to report.

4 A monitor on the real arrival stream

Everything above is a fixed-N exact test. The population is not fixed: the leaderboard and its successors add models continuously, and the natural downstream use is a standing monitor ("has correlated failure in the open ecosystem worsened since last quarter?") checked repeatedly as submissions arrive, without the type-I inflation that repeated testing on a growing sample incurs. That is the problem this community formalizes with e-values and test martingales (Shafer, 2021; Vovk & Wang, 2021; Ramdas et al., 2023; Grünwald et al., 2024). We build the version the exact null already supports, and report where it stops.

Construction. Order the ARC models by leaderboard submission date and take monthly looks over the accumulating population, N_t growing from 241 to 3,762 across twelve looks, on a snapshot pinned by content hash in the released artifacts (the archive is still growing, so a later snapshot gives different counts). At each look the null law of the accumulated matrix given its current margins is exactly uniform on a finite set, so the Monte Carlo p-value p_t for rms $|R_{ij}|$ against $R = 40$ curveball replicates drawn for that population is exactly valid. Calibrate each with $e_t = \kappa p_t^{\kappa-1}$, $\kappa = 1/2$ (Vovk & Wang, 2021), and merge by the arithmetic mean, which is an e-value under arbitrary dependence (Vovk & Wang, 2021) and so tolerates the nesting of the populations, which is as dependent as looks get. As a control we run the identical pipeline with the observed matrix at each look replaced by a curveball draw from that look's own null, a stream in which the null is true by construction.

What it shows. The control stream is the informative arm: its per-look p-values range over $[0.049, 0.951]$ and its merged e-value ends at 0.90 having never exceeded 1.17, so the construction does not accumulate evidence when there is none, which is the property that makes it worth calling a monitor. On real data every look returns the smallest p-value 40 replicates can express, so the merged e-value pins at 3.20; that number is the calibrator's value at $p = 1/41$ and is a statement about Monte Carlo resolution, not a likelihood ratio, and buying a larger one means buying more replicates rather than finding more evidence. The substantive reading is in the ratio column, flat to slightly declining as the population quadruples ($5.90\times$ at $N = 241$, $5.38\times$ at $N = 3{,}762$): residual co-failure is not visibly worsening over this window, and the monitor is what would show it if it were.

Look	N_t	rms ratio	p_t (real)	merged e (real)	merged e (control)
2023-07	241	$5.90\times$	0.024	3.20	0.51
2023-11	918	$5.58\times$	0.024	3.20	1.17
2024-01	1737	$5.24\times$	0.024	3.20	1.01
2024-03	2849	$5.36\times$	0.024	3.20	0.95
2024-06	3762	$5.38\times$	0.024	3.20	0.90

Table 2: Five of the twelve monthly looks. The control column is the same pipeline on a stream where the null holds by construction; its merged e-value never exceeds 1.17, which is the behaviour a valid e-value must show. The real stream sits at the Monte Carlo floor $p = 1/41$ at every look, so its calibrated e-value sits at that resolution's ceiling of 3.20.

What is still missing. This is valid for a pre-specified set of looks, not for optional stopping, which is what an e-process provides (Ramdas et al., 2023). A test martingale needs a null for the transition from the population at time t to that at $t + 1$, and conditioning does not supply one: given the first N_t rows and the new margins, the arriving row is determined, so there is no conditional law left to bet against. The sequential object must be built differently, for instance by conditioning each arriving row on its own row sum, which eliminates that model's ability parameter but not the item parameters, and handling the item parameters as a composite nuisance with a mixture or plug-in e-value, at the cost of the exactness that motivates the fixed-N test. A second obstacle is selection: models are submitted by people who observe the current leaderboard, so arrivals are not exchangeable, which is the population-dependence objection Jo et al. (2026) raise about a fixed null, posed sequentially. We bring both to this workshop as the open problem.

5 Limitations

Claims cover the evaluated open-model ecosystem on multiple-choice-style benchmarks, and the population is self-selected leaderboard submissions. The conditioning model is unidimensional; Section 3 rules out discrimination heterogeneity and duplicate models as sufficient explanations but cannot separate genuine behavioural concentration from an underfit null family, and on the spectral evidence the latter is the better description. Two candid points about the null: its rms sits only 1.25–$2.0\times$ above the analytic $1/\sqrt{M}$ floor, and conditioning on the item margins alone reproduces it to within 8.3% on every benchmark, so the item margins do essentially all of the conditioning work. Code, substrate, results artifacts and the dated pre-registration are archived at doi:10.5281/zenodo.21860005, `https://github.com/SilverCoin256/cofail` and `https://osf.io/cmh7q/`. A large language model assisted with code and drafting; the author pre-registered the hypotheses and kill conditions, verified every number against an executed-run artifact, and is responsible for the content.

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